

WORDPRESS EXAM REPORT

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Exam Question

Task: Manually Migrate a WordPress Site to a New Server

You are tasked with migrating a WordPress site from an old Debian 10 server running Apache to a new Debian 12 server, also running Apache. This must be done manually without the use of migration plugins. Below are the instructions and important details for completing the task.

Task Details:

- **First Zip:** Contains all WordPress site files (themes, plugins, uploads, etc.). Download it from [here](https://drive.google.com/file/d/1vw-ekfJ-eAjerilcjXwsQxlhSl8QnHTZ/view?usp=sharing) (<https://drive.google.com/file/d/1vw-ekfJ-eAjerilcjXwsQxlhSl8QnHTZ/view?usp=sharing>).
- **Second Zip:** Contains the exported WordPress database. Download it from [here](https://drive.google.com/file/d/1MsWTCgtkVYzDYf0TEmTavb_63ZCD6J9N/view?usp=sharing) (https://drive.google.com/file/d/1MsWTCgtkVYzDYf0TEmTavb_63ZCD6J9N/view?usp=sharing).

Instructions:

- Extract the WordPress site files from the first zip and transfer them to the new server.
- Import the database from the second zip file to the new server's database.
- Update the `wp-config.php` file to reflect the new database and server settings.
- Ensure all file and directory permissions are properly set.
- Test the site on the new server to confirm that everything works correctly (e.g., permalinks, plugins, themes).

Security Hardening for **lms.armour.local**:

- Implement SSL/TLS by obtaining and configuring an SSL certificate.
- Set appropriate file and directory permissions to secure the site.
- Configure a Web Application Firewall (WAF) to enhance security.
- Disable unnecessary features like XML-RPC and restrict REST API access for unauthorized users.
- Enable Two-Factor Authentication (2FA) for admin accounts.
- Set up real-time monitoring and alert systems to notify administrators of suspicious activities.
- Use IP blacklisting and geoblocking to prevent malicious traffic from specific regions or IP addresses.
- Restrict login attempts to prevent brute-force attacks.

Deliverables:

- A detailed configuration document (PDF) outlining the steps taken for migration, including the updated `wp-config.php` file (with sensitive information redacted).
- Screenshots or logs confirming the successful migration and testing of the site.

Assessment Criteria:

- Accurate extraction and transfer of files and database.
- Correct configuration of the WordPress site on the new server.
- Site functionality post-migration without errors.

Server Details

SERVER DETAILS	
OS	Debian 12
IPv4	192.168.169.164
Hostname	lms.armour.local
Domain	armour.local, lms.armour.local www.armour.local
Password	User – root, Password – root
Debian	User – Nikhil, Password – 123

Password Wordpress	User – admin, Password – ----- User – armour, Password – -----
Password Mysql	Mysql –u root –p, Password- root User - wp_user, Password – -----

~SERVER Basic Configuration

Machine Name Wordpress Exam Debian 12

Set IP and Gateway

nano /etc/network/interfaces

ip 192.168.169.164

gateway 192.168.169.135

Hostname Name Set

hostnamectl set-hostname lms.armour.local

reboot

Check

hostname

lms.armour.local

vim /etc/resolv.conf

Add this line

nameserver 8.8.8.8

nameserver 8.8.4.4

Disable IPv6

vim /etc/sysctl.conf

Add this Line

net.ipv6.conf.all.disable_ipv6 = 1

net.ipv6.conf.default.disable_ipv6 = 1

net.ipv6.conf.lo.disable_ipv6 = 1

Apply Changes Immediaelty

sysctl -p

~Apache Install & Setup

apt install apache2 -y

Service Enable, Restart and Status Check

systemctl enable apache2

systemctl start apache2

systemctl status apache2

Check Apache2

dpkg -l | grep apache2

~Install PHP on Debian 12

```
apt install php php-cli php-common php-opcache php-gd php-curl php-mysql php-xml php-mbstring php-pear
```

Check PHP Version

```
php -v
```

Restart Apache2 Service

```
systemctl restart apache2
```

Create phpinfo Test Page

```
nano /var/www/html/phpinfo.php
```

Add code

```
<?php phpinfo(); ?>
```

Then Open in Browser

```
http://192.168.169.164/phpinfo.php
```

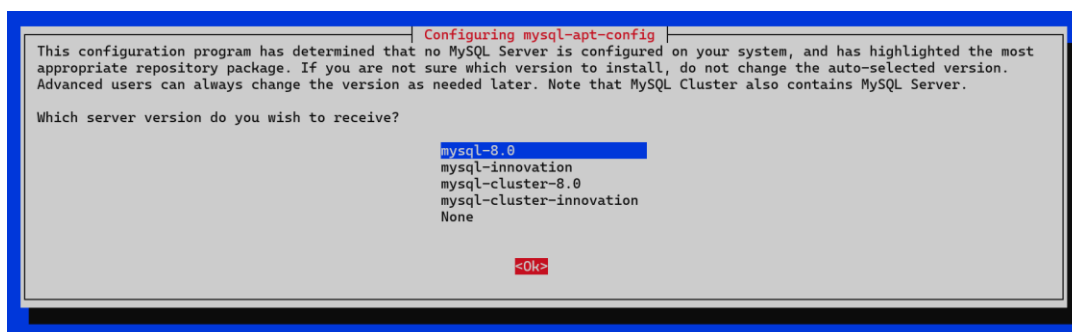
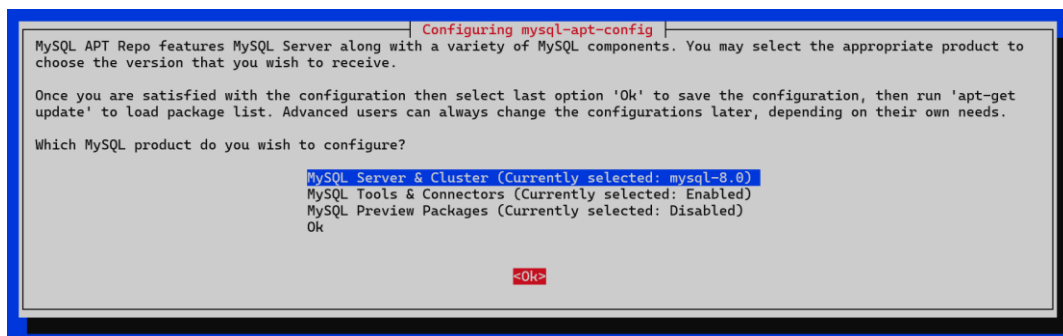
~MySQL Server install (official version)

```
apt install wget lsb-release gnupg
```

```
wget https://dev.mysql.com/get/mysql-apt-config_0.8.29-1_all.deb
```

```
dpkg -i mysql-apt-config_0.8.29-1_all.deb
```

```
apt install mysql-server
```



```
systemctl enable mysql
```

```
systemctl restart mysql
```

```
systemctl start mysql
```

```
mysql -u root -proot
```

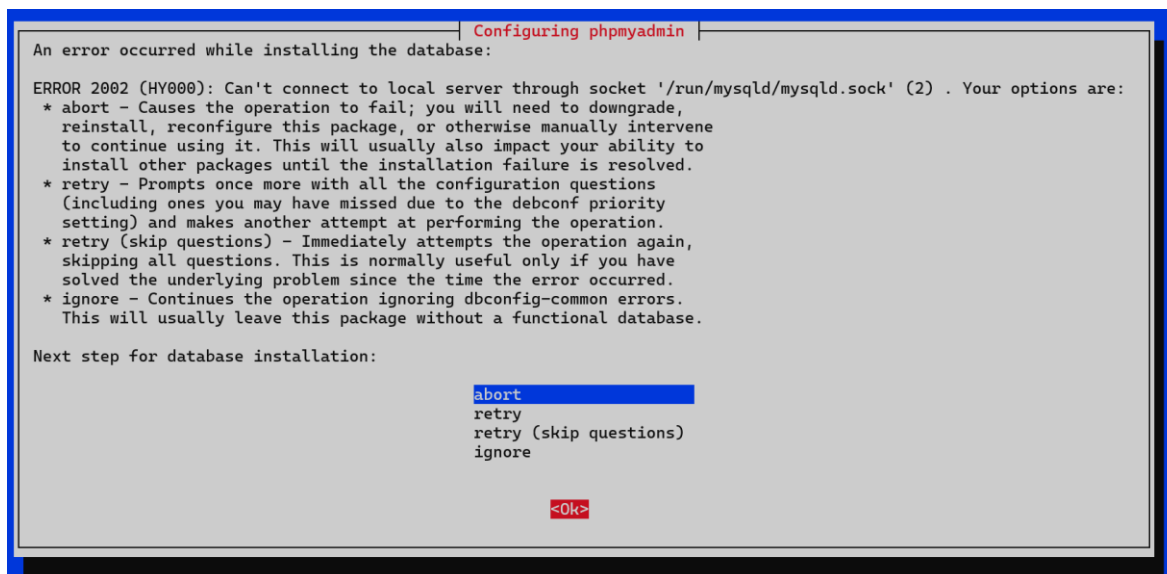
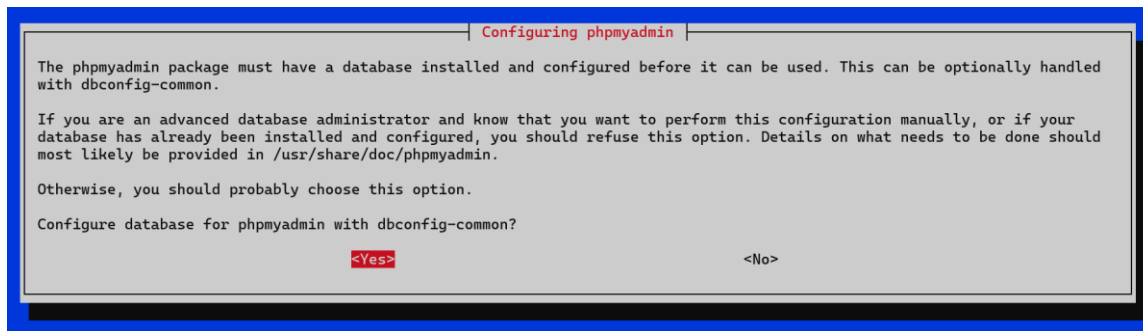
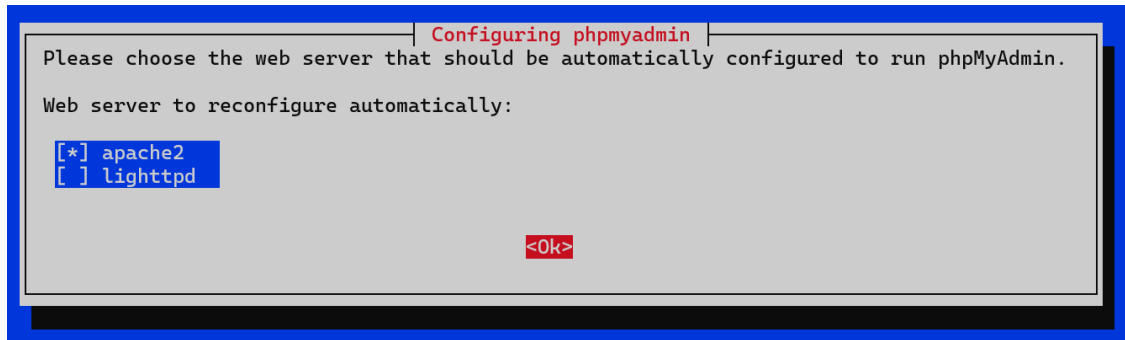
```
ALTER USER 'root'@'localhost' IDENTIFIED BY 'root';
```

```
SHOW DATABASES;
```

```
EXIT;
```

~Install phpMyAdmin

apt install phpmyadmin



Login PhpMyAdmin

Then Login PhpMyAdmin on Browser

<http://192.168.169.164/phpmyadmin>

Username – root

Password – root

~Create Database in Mysql

Open My MySQL

mysql -u root -proot

Create Database

CREATE DATABASE wordpress_db;

```
CREATE USER 'wp_user'@'localhost' IDENTIFIED BY '-----';  
GRANT ALL PRIVILEGES ON wordpress_db.* TO 'wp_user'@'localhost';  
FLUSH PRIVILEGES;  
EXIT;
```

Check Database

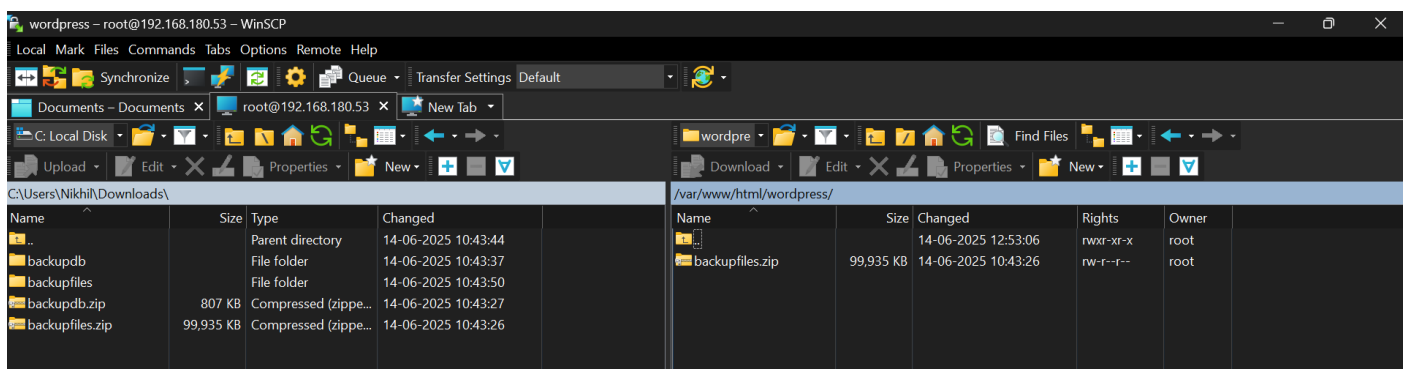
show Databases;

```
mysql> show Databases;  
+-----+  
| Database |  
+-----+  
| information_schema |  
| mysql |  
| performance_schema |  
| sys |  
| wordpress |  
+-----+  
5 rows in set (0.00 sec)
```

~Re-upload or re-extract the ZIP

```
scp backupfiles.zip root@192.168.169.164:/var/www/html/wordpress/
```

Windows Client Tools WinSCP



Then on the server:

```
cd /var/www/html/wordpress/
```

```
unzip backupfiles.zip
```

```
mkdir -p /var/www/html/wordpress
```

```
cp -r armour.local/* /var/www/html/wordpress/
```

Set Correct Ownership and Permissions

```
chown -R www-data:www-data /var/www/html/wordpress/
```

Update wp-config.php

```
vim /var/www/html/wp-config.php
```

Edit This Data

```
define('DB_NAME', 'wordpress_db');  
define('DB_USER', 'wp_user');  
define('DB_PASSWORD', '-----');  
define('DB_HOST', 'localhost');
```

Restart Apache and Test the Site

```
systemctl restart apache2
```

Open your browser and visit:

```
http://192.168.169.164/wordpress
```

~Create SSL Certificate

```
mkdir -p /etc/apache2/ssl
```

```
cd /etc/apache2/ssl
```

```
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/apache2/ssl/wordpress.key -out /etc/apache2/ssl/wordpress.crt
```

~Update Your Virtual Host for HTTPS

Open Configuration File

```
nano /etc/apache2/sites-available/lms.armour.local.conf
```

Add this Data

```
<VirtualHost *:443>
    ServerAdmin armour.local
    ServerName lms.armour.local
    DocumentRoot /var/www/html/wordpress

    SSLEngine on
    SSLCertificateFile /etc/apache2/ssl/wordpress.crt
    SSLCertificateKeyFile /etc/apache2/ssl/wordpress.key

    <Directory /var/www/html/wordpress>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/lms_error.log
    CustomLog ${APACHE_LOG_DIR}/lms_access.log combined

    # Optional: Extra security headers
    Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"
</VirtualHost>
```

Enable SSL & Headers Modules

```
a2enmod ssl
```

```
a2enmod headers
```

```
a2enmod rewrite
```

Enable the Site & Restart Apache

```
a2ensite lms.armour.local.conf
```

```
systemctl restart apache2
```

Update Local DNS (if not already)

```
C:\Windows\System32\drivers\etc\hosts
```

```
192.168.169.164 lms.armour.local
```

Update WordPress to Use HTTPS

MySQL -u root -p

```
USE wordpress_db;  
UPDATE wp_options  
SET option_value = 'https://lms.armour.local'  
WHERE option_name IN ('siteurl', 'home');
```

Update configuration file

nano wp-config.php

Add

```
define('WP_HOME', 'https://lms.armour.local');  
define('WP_SITEURL', 'https://lms.armour.local');
```

Finnal Run Website

https://lms.armour.local/



~Install DNS Required Packages

```
apt install bind9 dnsutils
```

Create Zone

```
vim /etc/bind/named.conf.local
```

```
zone "armour.local" {  
    type master;  
    file "/etc/bind/zones/forward.armour.local";  
};
```

Create Forward Zone Record

```
mkdir /etc/bind/zones  
vim /etc/bind/zones/forward.armour.local
```

Add this Data

\$TTL 604800

```
@ IN SOA lms.armour.local. root.armour.local. (  
    3      ; Serial  
    604800 ; Refresh  
    86400  ; Retry  
    2419200 ; Expire  
    604800 ) ; Negative Cache TTL
```

; Nameserver declaration

```
@ IN NS lms.armour.local.
```

; A records (host to IP mappings)

```
@ IN A 192.168.169.164  
ns IN A 192.168.169.164  
lms IN A 192.168.169.164
```

Host Entry

Vim /etc/hosts

Add this Line

192.168.169.164 armour.local lms.armour.local www.armour.local

Restart DNS Service

systemctl restart bind9

systemctl enable named.service

Check Resolv

nslookup lms.armour.local 192.168.169.164

~Reset Password Wodpress User and Admin

mysql -u root -p

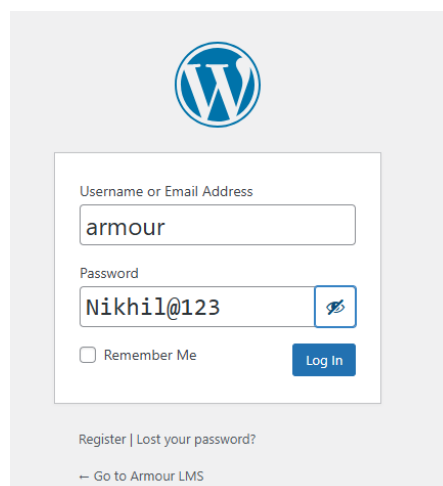
USE wordpress_db;

SELECT ID, user_login FROM wp_users;

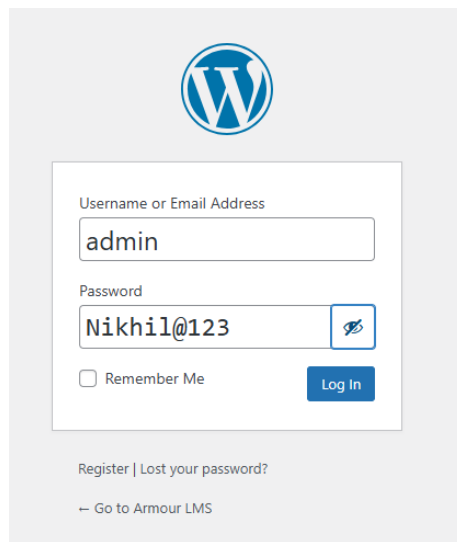
```
mysql> SELECT ID, user_login FROM wp_users;  
+----+-----+  
| ID | user_login |  
+----+-----+  
| 1  | admin      |  
| 2  | armour     |  
+----+-----+  
2 rows in set (0.01 sec)
```

UPDATE wp_users SET user_pass = MD5('-----') WHERE user_login = 'armour';

UPDATE wp_users SET user_pass = MD5('-----') WHERE user_login = 'admin';



The image shows a WordPress login page for a site named 'Armour LMS'. At the top is the WordPress logo. Below it is a login form with two input fields: 'Username or Email Address' containing the text 'armour' and 'Password' containing 'Nikhil@123'. There is a 'Remember Me' checkbox which is unchecked, and a 'Log In' button. At the bottom of the form, there are links for 'Register' and 'Lost your password?'. Below the form, there is a link that says 'Go to Armour LMS'.



~Web Application Firewall (WAF) — Wordfence Setup

(SQL Injection, XSS, Brute Force Attack, File Upload Exploit, DDOS Attacke,)

Go to: <https://lms.armour.local/wp-admin>

Plugins → Add New

Search: Wordfence

Click Install → Activate

☐	Wordfence Security Upgrade To Premium Deactivate	Wordfence Security - Anti-virus, Firewall and Malware Scan Version 8.0.5 By Wordfence View details	Enable auto-updates
--------------------------	--	---	-------------------------------------

Check

[https://lms.armour.local/?author\[\]=1' OR 1=1](https://lms.armour.local/?author[]=1' OR 1=1)—

error 404 Page

~Disable XML-RPC

(Brute Force Login, DDOS via PingBacks, Botnet Coordination)

Plugin: Disable XML-RPC

Go to: <https://lms.armour.local/wp-admin/plugins.php>

Search for: Disable XML-RPC

Click: ⓘ Install → ⓘ Activate.

Check

<https://lms.armour.local/xmlrpc.php>

Forbidden

~Restrict REST API Access

Open Your WordPress Theme Directory

`cd /var/www/html/wordpress/wp-content/themes/`

Choose Active Theme

`cd twentytwentyfour`

Open `functions.php`

nano functions.php

Add this Data

```
// Completely block REST API access for non-logged-in users
add_filter( 'rest_authentication_errors', function( $result ) {
    if ( ! is_user_logged_in() ) {
        return new WP_Error(
            'rest_cannot_access',
            __( 'REST API is restricted to authenticated users only.' ),
            array( 'status' => 403 )
        );
    }
    return $result;
});
```

Check

<https://lms.armor.local/wp-json/wp/v2/users>

```
{"code":"rest_user_cannot_view","message":"Sorry, you are not allowed to list users.","data":{"status":401}}
```

~Real-time Monitoring (Wordfence)

(Attack Detection, Email Alerts, Live IP Tracking, Threat Blocking, Scheduled Malware Scans)

Live Traffic Dashboard

Wordfence → Tools → Live Traffic

Wordfence → All Options → Email Alert Preferences

Wordfence → Scan → Scan Options

Wordfence → Tools → Diagnostics → Enable Debugging Mode

~IP Blacklisting / Geo-Blocking

(Block IPs, Geo-Blocking, Block User Agents, Custom Patterns)

Open Configuration file

nano /var/www/html/wordpress/.htaccess

Add at the Very Top:

Block Specific IPs

<Limit GET POST>

Order allow,deny

Deny from 192.168.169.144

Deny from 23.45.67.89

Allow from all

</Limit>

Result

<https://lms.armor.local/wp-admin/>



Forbidden

You don't have permission to access this resource.

Apache/2.4.62 (Debian) Server at lms.armor.local Port 443

~Limit Login Attempts (Brute Force Protection)

Go to: Plugins

Plugins → Add New

Search: Limit Login Attempts Reloaded

Click Install → Activate

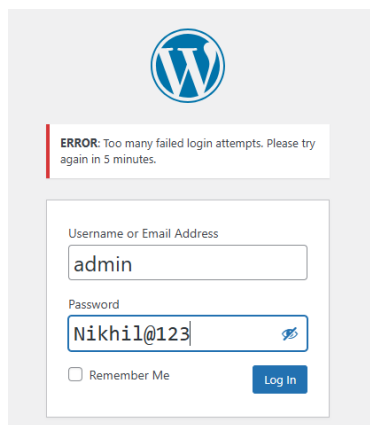
Configure Settings

Settings → Limit Login Attempts Reloaded

Option	Value
Lockout after how many retries	5
Lockout time (minutes)	60
Hours until retries are reset	4
Lockouts increase after X attempts	4
Show message on lockout?	☒ Yes
Notify on lockout via email?	☒ (Optional)

Check

<https://lms.armour.local/wp-admin/>



~Enable 2FA (Two-Factor Authentication)

Install the Plugin

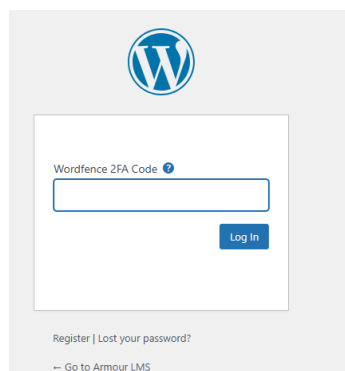
Go to: WP Admin > Plugins > Add New

Search: Wordfence Login Security

Click Install Now

Then Activate

Barcode scan on Authy mobile application and submit successfully Two Factor Authentication Enable



MySQL queries to update WordPress URLs after migrating or changing domains.

Open Login MySQL

Logs into MySQL with the root user. You'll be prompted for the password.

```
mysql -u root -p
```

Switches to the WordPress database named wordpress_db.

```
USE wordpress_db;
```

Replaces old domain links inside post content (pages, posts, custom content).

```
UPDATE wp_options SET option_value = 'https://lms.armour.local' WHERE option_name = 'siteurl';
```

```
UPDATE wp_options SET option_value = 'https://lms.armour.local' WHERE option_name = 'home';
```

Updates the main WordPress site URLs to the new domain <https://lms.armour.local>.

```
UPDATE wp_posts SET post_content = REPLACE(post_content, 'https://www.armour.local',  
'https://lms.armour.local');
```

```
UPDATE wp_posts SET guid = REPLACE(guid, 'https://www.armour.local', 'https://lms.armour.local');
```

Replaces old GUIDs. Be cautious: normally GUIDs shouldn't be changed unless you're changing domains.

Catches any remaining links in serialized or plugin data inside wp_options.

```
UPDATE wp_options SET option_value = REPLACE(option_value, 'https://www.armour.local',  
'https://lms.armour.local');
```

Pro Tip:

If you have widgets, plugin settings, or serialized data with URLs, consider using a search-replace tool like:

```
wp search-replace 'https://www.armour.local' 'https://lms.armour.local' --skip-columns=guid  
(Requires WP-CLI installed)
```

Final Checklist:

- ☐ Site loads correctly at new domain
 - ☐ Media/image links work
 - ☐ Dashboard/login works at <https://lms.armour.local/wp-admin>
 - ☐ Permalinks updated (optional: visit Settings > Permalinks > Save)
-

Perfect, Nikhil! What you did was **the complete professional-grade method** for updating a WordPress domain when Elementor is in play. Here's a clean **summary of your steps** for documentation, viva prep, or audits:

☐ WordPress Domain Change with Elementor (GUI Method)

1. ☐ Logged into WP Admin

→ Gained access to the backend via <https://lms.armour.local/wp-admin>.

2. ☐ Elementor → Tools → Replace URL

→ Replaced:

3. Old URL: <http://www.armour.local>

4. New URL: <https://lms.armour.local>

☐ Elementor safely updated all URLs across pages, posts, widgets, templates, and **serialized meta fields**.

5. ☐ **Regenerated Elementor Files**
→ Went to **Elementor** → **Tools** → **General Tab**
→ Clicked **“Regenerate Files & Data”**
☐ Ensures all cached styles, scripts, and content paths are rebuilt correctly.
 6. ☐ **Flushed Permalinks**
→ Navigated to **Settings** → **Permalinks**
→ Clicked **“Save Changes”**
☐ Forces WordPress to rebuild `.htaccess` rules and fix any broken permalinks.
-

Bonus Recommendation:

After all this, **clear all caches**:

- ☐ Browser cache
 - ☐ Elementor cache
 - ☐ WordPress object/cache plugin (if used)
 - ☐ CDN cache (Cloudflare, etc.)
-

Why This Method Is Gold?

- ☐ **Safe with serialized data** (unlike raw SQL replace).
 - ☐ Prevents breaking widget templates or page builder layouts.
 - ☐ Rebuilds Elementor’s cache files to prevent CSS/JS loading issues.
 - ☐ Plugin-friendly — doesn't touch unrelated tables or GUIDs unless necessary.
-
-

Wordpress Domain Change with Elementor (GUI Method)

1. Logged into WP Admin
→ **Gained access to the backend via `https://lms.armour.local/wp-admin`.**
2. Elementor → Tools → Replace URL
→ **Replaced:**
3. **Old URL:** `http://www.armour.local`
4. **New URL:** `https://lms.armour.local`

 **Elementor safely updated all URLs across pages, posts, widgets, templates, and serialized meta fields.**

5. Regenerated Elementor Files
→ **Went to** Elementor → Tools → General Tab
→ **Clicked** “Regenerate Files & Data”
 **Ensures all cached styles, scripts, and content paths are rebuilt correctly.**
 6. Flushed Permalinks
→ **Navigated to** Settings → Permalinks
→ **Clicked** “Save Changes”
 7.  **Forces WordPress to rebuild `.htaccess` rules and fix any broken permalinks.**
-

Report Completed

All the necessary components of the Wordpress environment have been configured and documented. This marks the successful completion of the Website Migration exam report.
